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# Clinical outcomes following inpatient penicillin allergy testing: A systematic review and meta-analysis

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## Abstract

**Background:** A documented penicillin allergy is associated with increased morbidity including length of hospital stay and an increased incidence of resistant infections attributed to use of broader-spectrum antibiotics. The aim of the systematic review was to identify whether inpatient penicillin allergy testing affected clinical outcomes during hospitalization.

**Methods:** We performed an electronic search of Ovid MEDLINE/PubMed, Embase, Web of Science, Scopus, and the Cochrane Library over the past 20 years. Inpatients having a documented penicillin allergy that underwent penicillin allergy testing were included.

**Results:** Twenty-four studies met eligibility criteria. Study sample size was between 24 and 252 patients in exclusively inpatient cohorts. Penicillin skin testing (PST) with or without oral amoxicillin challenge was the main intervention described (18 studies). The population-weighted mean for a negative PST was 95.1% [CI 93.8-96.1]. Inpatient penicillin allergy testing led to a change in antibiotic selection that was greater in the intensive care unit (77.97% [CI 72.0-83.1] vs 54.73% [CI 51.2-58.2], $P<.01$). An increased prescription of penicillin (range 9.9%-49%) and cephalosporin (range 10.7%-48%) antibiotics was reported. Vancomycin and fluoroquinolone use was decreased. Inpatient penicillin allergy testing was associated with decreased healthcare cost in four studies.

**Conclusions:** Inpatient penicillin allergy testing is safe and effective in ruling out penicillin allergy. The rate of negative tests is comparable to outpatient and perioperative data. Patients with a documented penicillin allergy who require penicillin should be tested during hospitalization given its benefit for individual patient outcomes and antibiotic stewardship.

KEYWORDS
antibiotic utilization, drug allergy, hospitalization, inpatient, penicillin testing

## 1 | INTRODUCTION

Drug allergy is a term which by convention describes immunologically mediated drug hypersensitivity reactions.[1,2] Drug allergy may manifest in any form of early or delayed hypersensitivity as outlined by the Gell and Coombs classification.[3] A true drug allergy is unpredictable and associated with significant morbidity and healthcare utilization.[4,5] Nonallergic drug reactions may mimic drug allergy leading to mislabeling patients with a drug allergy. Such patients receive less effective and more expensive pharmacotherapy.[5] Penicillin allergy is the most common drug allergy documented in a medical record.[6,7] The prevalence of reported penicillin allergy is estimated between 9% and 12%.[8,9] The prevalence may be as high as 15% in hospitalized patients.[10] A majority of patients with a history of

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penicillin allergy have no subsequent reactions to penicillin or beta-lactam antibiotics (BLAs).[5] It is thought that up to 90% of these patients have an unnecessary label of "penicillin allergy".[5] This is partly due to overuse of the term "allergy" when confounding symptoms related to the patient's illness may be attributed to penicillin "allergy".[1] However, such patients commonly receive second-line broad-spectrum antibiotics for infections that may be treated with first-line BLAs.[11] Common antibiotics that substitute first-line penicillins include fluoroquinolones, clindamycin, vancomycin, and third-generation cephalosporins. For this reason, a documented penicillin allergy increases patient risk for infections caused by *Clostridium difficile*, methicillin-resistant *Staphylococcus aureus* (MRSA), and vancomycin-resistant *enterococcus* (VRE).[6] Furthermore, a documented penicillin allergy may be associated with a prolonged hospital stay related to longer duration of treatment and adverse effects related to second-line antibiotic use.[6] This may have implications on healthcare utilization and cost.

A significant number of hospitalized patients are treated with antibiotics.[12] Hospitalization offers an opportunity to address and rule out a penicillin allergy especially as adverse outcomes related to a penicillin allergy are common during a hospitalization event. Penicillin skin testing (PST) is a validated tool to test for IgE-mediated penicillin hypersensitivity. Drug challenge is, however, the gold standard to rule out any form of drug hypersensitivity.[2] PST has a very high negative predictive value when both major and minor determinants are used.[2,13] BLAs can thus be safely administered following a negative PST.[14,15] Despite some initial concerns that an inpatient PST may not be valid due to interfering inpatient factors, PST for major and minor determinants was negative in 78% of 726 hospitalized patients with a history of penicillin allergy.[16] This value was comparable to other prior studies.[17,18] However, there has been a steady decline in the proportion of PSTs since 1995 that has been independent of study subject's age and time since reported penicillin reaction.[12] Data over the past 20 years have consistently shown the proportion of negative PSTs to be between 95% and 99% in outpatient, perioperative, and inpatient settings.[19] However, studies of inpatient populations have had small cohorts when compared to Sogn et al., 1992, calling for an updated negative PST in a larger inpatient population.[16]

The Infectious Diseases Society of America and Society for Healthcare Epidemiology of America guidelines for implementing an antibiotic stewardship program determined that penicillin skin testing to assess penicillin allergy merits a weak recommendation.[20] However, there is substantial morbidity related to unconfirmed penicillin allergy.[21] Multiple studies have consistently shown inpatient PST to decrease use of broad-spectrum antibiotics and decrease cost.[11,22,23] However, inpatient PST (with or without oral amoxicillin challenge) is not often readily available with only around 60% of institutions having PST available.[24]

Other inpatient interventions to address a penicillin allergy may include pharmacy-led initiatives or guideline-driven protocols with administration of penicillin test doses in patients deemed to be low risk for anaphylaxis.[10,25] In such cases, patient risk is stratified based on history and nature of the prior presumed penicillin allergy. Concerns for such an approach are that patients with a vague history of penicillin allergy may have a positive PST.[26] It is as yet unclear whether inpatient interventions addressing a penicillin allergy have any implications on clinical outcomes other than antibiotic utilization and cost. Any inpatient intervention that would impact patient morbidity (such as length of hospitalization, discharge on intravenous antibiotics, and readmission rate) may increase rates of inpatient allergy testing and make a case for improving inpatient infrastructure for penicillin allergy testing. This systematic review and meta-analysis was designed to address the following aims:

- To outline major inpatient interventions that address a penicillin allergy during an index admission when a patient would require a penicillin antibiotic.
- To ascertain the validity of inpatient penicillin skin testing to safely rule out a penicillin allergy.
- To identify clinically relevant outcomes following inpatient penicillin allergy testing.

## 2 | METHODS

### 2.1 | Study criteria required for systematic review

#### 2.1.1 | Types of studies

We included case-cohort studies, interventional cohort studies, and randomized controlled trials that included an inpatient intervention to ascertain penicillin allergy in patient admitted with documented penicillin (PCN) allergy on their medical record. The intervention had to be performed during the index admission. Studies of pediatric populations (age <18 years) and perioperative or obstetric populations were excluded.

#### 2.1.2 | Types of interventions

We included and evaluated studies that implemented any intervention to rule out a penicillin allergy whether it was a consult service-driven process, guideline-driven protocol, or standard (PST) for major and minor determinants with or without oral penicillin challenge.

#### 2.1.3 | Types of outcome measures

We included all outcomes described including the following: adverse outcomes related to the inpatient intervention, proportion of patients testing negative for PCN allergy, change in management including antimicrobial administration, adverse outcomes related to change in management following testing, effect on cost, and other longitudinal healthcare outcomes.

### 2.2 | Search methods

An experienced medical research librarian developed and ran searches in the MEDLINE, Embase, Cochrane Database of

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Systematic Reviews, Cochrane Central Register of Controlled Trials (all via the Ovid interface), Web of Science, and Scopus databases. The search strategies were peer-reviewed by another experienced librarian and revised prior to finalization. Search terms included MeSH and Embase terms as well as keywords including hospitalization, inpatients, drug hypersensitivity, allergy, penicillins, and beta-lactams. Limitations to the search strategies included the inclusion of only English-language publications and a publication date range of the last 15 years. A total of 2212 citations were retrieved from the six databases. After removal of duplicates, total number of citations left equaled 1836. All database searches were run on December 6, 2016. The full search strategies are available in the Appendix S1. Abstracts, conference proceedings, and cross-checked references of selected studies were also reviewed by authors to identify additional studies that had not been identified by database searches.

### 2.2.1 | Data collection and analysis

T.J.B. conducted the search strategy and screened for and identified duplicate titles and abstracts from the database searches. Two review authors (K.A.S and A.B.) independently screened the abstracts using prepiloted screening forms. Any disagreements were resolved by consensus or by consulting a third review author (M.C.B.). Both full-text manuscripts and conference abstracts at national or international level were considered for inclusion. The remaining studies were evaluated as full-text publications using predetermined inclusion and exclusion criteria. Reason for exclusion of each study was documented at this stage. We recorded the study design, sample size, sampling methods, type of inpatient population, inpatient intervention addressing penicillin allergy, and subsequent clinical outcomes described. We contacted the study authors to obtain additional information if it was not available from the published reports or reports from the studies. We conducted a qualitative analysis of the study cohort and inpatient intervention to address PCN allergy and assessed the clinical outcomes described.

We assessed the risk of bias in included studies using the Newcastle-Ottawa Quality Assessment Scale for cohort studies and case-control studies (Table S2 and S3).[27] We performed population-weighted analysis on all comparable observational cohort studies reporting the rate of negative PSTs to assess whether it would be a reproducible test when compared to outpatient data. We calculated the 95% confidence interval for population-weighted means using the Wilson score interval.[28] We performed population-weighted analysis on cohort studies reporting a common clinical outcome after implementation of the same or consistently similar inpatient penicillin allergy test.

## 3 | RESULTS

Our search strategy identified 1836 unique bibliographic citations of which 74 were selected for full-text evaluation (Figure 1). We identified 24 studies, 17 full-text articles, and seven conference abstracts. Two studies were case-cohort studies, while the other 22 studies were cohort studies.[10,11,15,22,23,25,29-46] The included studies were grouped according to their inpatient population, inpatient intervention, and clinical outcomes described (Figure 2). Table S1 summarizes the study characteristics, sample size, and study design. The inter-rater agreement between the two review authors (K.A.S. and A.B.) was 89.2% when selecting studies for clinical outcomes with a Cohen's kappa of 0.44 (CI 0.13-0.76).

Six studies had patient cohorts exclusively in the intensive care unit (ICU),[22,29-32,38] and one study had a mixed ICU and general inpatient population.[37] Del Real et al. reported outcomes of PST in a heterogeneous inpatient and outpatient population (n=596) of which 50.3% were inpatient and 24.3% were admitted to the ICU.[15] Sample size varied between 24 and 146 subjects in exclusively inpatient cohorts testing for penicillin allergy. Blumenthal et al., 2015, reported an increase in BLA test doses and decrease in vancomycin and quinolone use following an inpatient guideline to rule out penicillin allergy.[32] The study was omitted from pooled-cohort analysis as the primary outcome was to assess increase in BLA test doses rather than individual patient or population-based clinical outcomes. Specific studies described specific infection groups such as infective endocarditis[45] and MSSA bacteremia.[33]

The commonest inpatient intervention performed was penicillin skin prick or intradermal testing with or without subsequent challenge of oral penicillin (or its derivative). This was the primary intervention in 19 studies (Table S1, Figure 2). The proportion of negative PSTs ranged between 79% and 100%. Motosue et al. reported a negative PST in 79% of inpatients with 20% of tests being indeterminate.[40] Given that the study was a retrospective review, patients with indeterminate tests could not be retested. The population-weighted mean for a negative PST was 95.1% [CI 93.8-96.1] (Figure 3). One study implemented an inpatient allergy consult prior to PST.[43] Sigona et al., 2016, reported a pharmacy-driven patient interview that showed 65.6% of patients were subsequently switched to BLAs.[25]

The most commonly reported clinical outcome was a change in antimicrobial therapy (20 studies). Studies consistently reported change to a narrower spectrum antibiotic, switching to BLAs, and subsequently increased prescription of penicillin and cephalosporins. Studies consistently reported a decrease in vancomycin and fluoroquinolone use following PCN allergy testing (Figure 2). A pooled-cohort analysis showed a change in antibiotic prescription of 77.97% [CI 72.0-83.1] in the ICU setting with a change to a lesser proportion (54.73% [CI 51.2-58.2]) in other inpatient studies (Figure 4).

Two studies reported an annual healthcare saving,[22,37] three studies found a cost saving per individual patient,[11,23,42] while two studies were unable to report a statistically significant cost benefit[31,35] (Table S1, Figure 2).

Very few adverse events related to testing were reported. These were individual cases of anaphylaxis that were more common with oral amoxicillin challenge at an incidence of less than 1 percent.[11,15,23,30,37]

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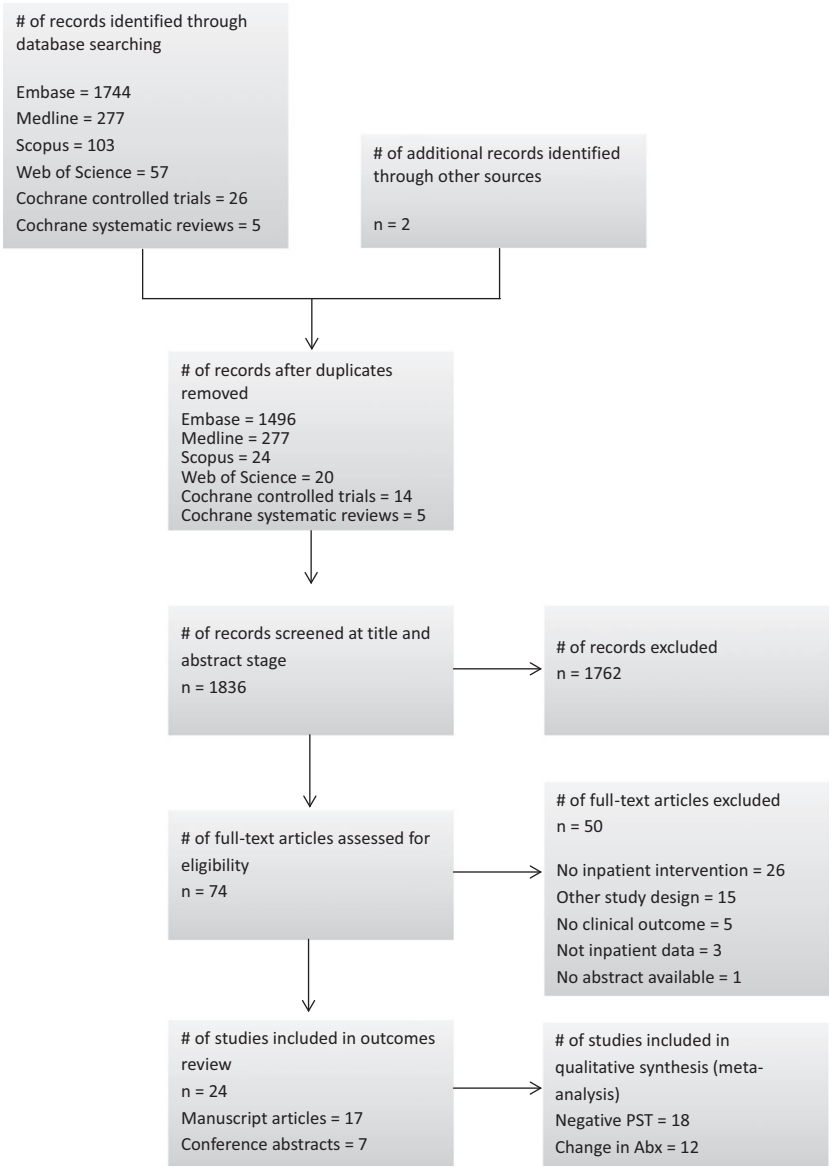


**FIGURE 1** Study flow diagram. PST=penicillin skin test, Abx=antibiotics, BLA=B-lactam antibiotic

## 4 | DISCUSSION

PST is a safe and reproducible test to rule out immunoglobulin E (IgE)-mediated penicillin allergy in the inpatient setting. The negative predictive value of inpatient allergy testing is comparable to outpatient and perioperative cohort data.[19,47] Furthermore, the intervention has consistently shown to reduce the use of broad-spectrum antibiotics, a finding that has been reported from studies in the perioperative setting.[48] Antibiotic prescription changes in the inpatient setting may have greater implications than in the perioperative setting. Most inpatients receive antibiotics with therapeutic intent rather than prophylaxis with longer treatment duration. Further, the surgical population is likely to have less comorbidity than a medical inpatient population. It was reassuring to note that adverse events to PST were few with most studies reporting no adverse reactions. An isolated case of self-limiting hives was reported following intradermal PST.[37] One case of anaphylaxis was reported following second dose of graded oral amoxicillin challenge that was treated with epinephrine and antihistamines.[11] No permanent morbidity or death occurred in any study with inpatient PST and oral antibiotic challenge. Of note, however, most studies failed to classify reactions as early or delayed hypersensitivity reactions.

PST is a bioassay that detects the presence of allergen-specific IgE on a patient's mast cells. A transient "wheal-and-flare" is noted over the skin surface within 15-20 minutes of allergen application, characteristic of a Gell and Coombs type 1 hypersensitivity.[19] Skin testing is the fastest and most cost-effective validated test to evaluate adult and pediatric patients with a presumed allergy to penicillin and similar drug classes.[19] However, it must be reiterated that a PST in isolation does not rule out the spectrum of immunologically mediated hypersensitivities that encompass drug allergy. PST was traditionally reserved for patients with no alternative other than penicillin for their infection. However, PST has emerged to play a pivotal role

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| Population | Intervention | Outcome described | Effect | Study |
|---|---|---|---|---|
| Inpatient | Penicillin skin test | Overall change in antibiotic prescription | 38% change in prescription | Chen et al., 2016 [36] |
| | | | 68% change in prescription | Heil et al, 2016 [38] |
| | | | 78.9% change in prescription | Ward et al., 2015 [47] |
| | | | 57.4% change to β-lactam | Del Real et al., 2007 [15] |
| | Pharmacy-driven interview | | 65.6% change in prescription | Sigona et al., 2016 [26] |
| | Penicillin skin test | Change in antibiotic class | Increase in penicillin | Macy et al., 2004 [41] |
| | | | | Nadarajah et al;, 2005 [43] |
| | | | | Perencevich et al., 2001 [46] |
| | | | Increase in cephalosporin | Macy et al., 2004 [41] |
| | | | | Nadarajah et al., 2005 [43] |
| | | | | Perencevich et al., 2001 [46] |
| | | | Decreased vancomycin | Macy et al., 2004 [41] |
| | | | | Nadarajah et al., 2005 [43] |
| | | | | Perencevich et al., 2001 [46] |
| | | | Decreased fluoroquinolone | Nadarajah et al., 2005 [38] |
| | | | | Perencevich et al., 2001 [41] |
| | | Cost saving | Yes | King et al., 2016 [11] |
| | | | No | Forrest et al., 2001 [37] |
| | | Persistence of allergy on record | 10% | Chen et al., 2016 [36] |
| | | | 49% | Warrington et al., 2000 [48] |
| ICU | Penicillin skin test | Change in antibiotic prescription | 48% change in prescription | Arroliga et al, 2000 [31] |
| | | | 81.5% change in prescription | Arroliga et al, 2003 [30] |
| | | | 70.3% change in prescription | Del Real et al., 2007 [15] |
| | | | 100% change in prescription | Rimawi et al., 2014 [40] |
| ICU and inpatient | Penicillin skin test | Change in antibiotic prescription | 100% in all test-negative | Rimawi et al., 2013 [39] |
| | | Cost saving | Yes | Rimawi et al., 2013 [39] |

**FIGURE 2** Description of study interventions and outcomes described. Only full manuscript studies were included. ICU: intensive care unit

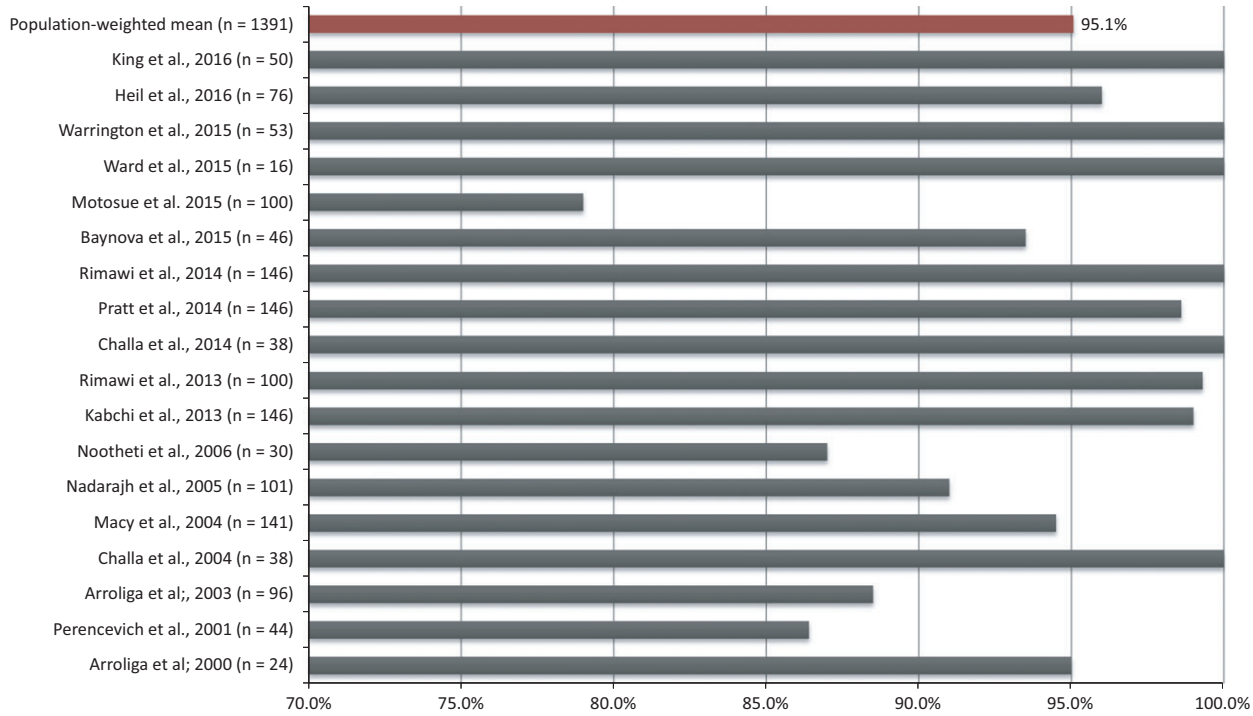


**FIGURE 3** Proportion of inpatient negative penicillin skin tests (PSTs) ranging between 79% and 100% in studies of exclusively inpatient (including intensive care unit) populations. Population-weighted mean of a negative PST in an inpatient setting is 95.1% (CI 93.8-96.1)

in antibiotic stewardship to avoid unnecessary use of broad-spectrum antibiotics that may be less effective, more toxic, more expensive; require lengthier intravenous administration; and are associated with prolonged hospital stay.[5,6] For the aforementioned reasons, a history of presumed penicillin allergy is not as benign as previously thought.

Identifying changes in antibiotic prescription is an easily measurable outcome that is suggestive of decreased adverse effects

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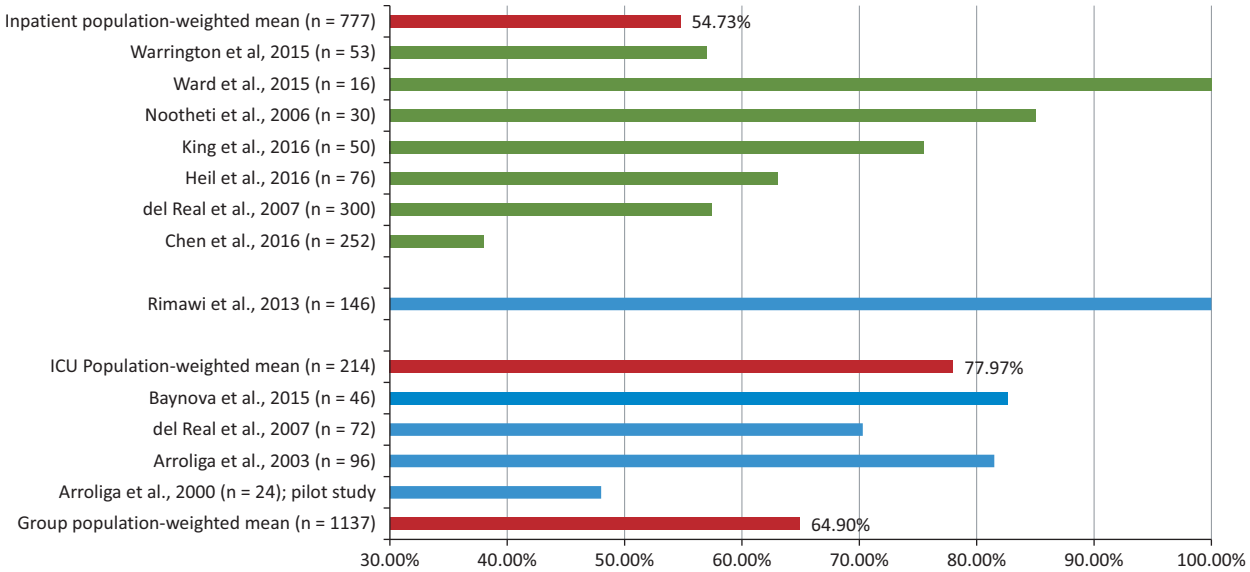


**FIGURE 4** Cohort studies that quote a change in antimicrobials during index admission following negative inpatient penicillin skin testing (PST). Perencevich et al.,[44] report 86% of PST-negative patients receiving beta-lactam antibiotics in subsequent admissions. Case-cohort studies were excluded together with studies that did not report overall change in antimicrobials. Results of exclusively inpatient populations excluding the intensive care unit (ICU) shown in green; results of exclusively ICU patients in blue; study by Rimawi et al.,[37] includes both ICU and inpatient services. The population-weighted mean for change in antimicrobials in the inpatient setting (excluding ICU) was 54.73% (CI 51.2-58.2), while a percentage change of 77.97% (CI 72.0-83.1) is seen in the ICU cohort. The total population-weighted mean for both groups is 64.90% (CI 62.1-67.6). Pilot studies were excluded from population-weighted means

associated with use of broad-spectrum antibiotics. However, this may not be necessarily so given that broad-spectrum antibiotics may be prescribed during subsequent admissions particularly in patients who had a negative inpatient PST but did not have their penicillin allergy removed from their medical record. Three studies reported 22.23%,[43] 49%,[46] and 92%[42] of patients having the allergy still listed on the chart following a negative test. The magnitude of this lamentable statistic may be underestimated given that it may have been overlooked as a significant clinical outcome. The heterogeneity of results suggests that the persistence of a penicillin allergy on a medical record may be affected by local hospital practices and ease of navigation within the allergy section of the electronic medical record. However, it highlights a gross missed opportunity for subsequent healthcare encounters and may overestimate the benefit of changes in antibiotics during the index admission.

A substantial change was noted in prescribing narrower spectrum antibiotics following PST (Figure 3). A rise in both penicillin and cephalosporin antibiotics was seen, possibly owing to the concern for cross-reactivity with cephalosporins in a history of penicillin allergy.[49] Conversely, a decrease in vancomycin and fluoroquinolone use was reported in all studies assessing this change. This suggests a possible relative risk reduction in acquiring health care-related infection such as VRE and *C. difficile*, albeit such outcomes were not measured directly possibly owing to small patient cohorts. A greater change in antibiotic prescribing practices was noted in the ICU setting as compared to the general inpatient population following penicillin testing (Figure 4). This occurred despite comparable negative PSTs in both cohorts. Such findings may be related to ICU physician-related awareness and need to rule out penicillin allergy when dealing with complex patients likely having multi-organ dysfunction. Furthermore, the growing problem of antibiotic resistance in the ICU is associated with greater mortality, morbidity, and healthcare cost.[50]

While PST is a valid and sensitive test to rule out an immediate hypersensitivity to penicillin, it is worth noting to a drug challenge remains the gold standard to rule out a drug hypersensitivity reaction.[1] Further, European populations have a lower rate of reactivity to the major determinant penicilloyl-polylysine in a standard PST. This is likely due to the higher prevalence of amoxicillin/ampicillin allergy particularly in Southern Europe with patients being sensitized to the R-group side chain. Populations in the United States are more likely to develop hypersensitivity to the beta-lactam core.[51] As a result, adverse events may occur at a higher rate in European populations if only PST is used. Owing to such geographical differences in sensitization profiles, a standard evaluation for penicillin allergy should include a PST followed by graded drug challenge when appropriate.[52,53] This would also follow consensus guidelines on evaluating drug allergy as one may miss immunologically mediated hypersensitivities other than IgE-mediated reactions using PST alone.[1]

A longitudinal follow-up with age- and sex-matched controls would be helpful to delineate specific clinical outcomes following PST that may be related to use of broad-spectrum antibiotics. Specifically, length of hospitalization, risk of health care-associated infection (MRSA, VRE, and *C. difficile* infection), and readmission rate should be analyzed as well as other clinically relevant longitudinal

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outcomes. The inpatient population is generally older, having greater comorbidities to the outpatient and perioperative cohorts. Further, a greater prevalence of penicillin allergy is described in the inpatient setting.[16,49,54,55] The studies identified in this review possibly underestimate the impact of inpatient PST given that most outcomes were related to the index admission. Sustained use of narrower spectrum antibiotics has been shown to persist during readmissions.[11,35] However, a number of questions related to patient demographics remain unanswered. It is not known whether specific patient, disease, or hospital characteristics could affect which populations got testing. Further, it is imperative to know the timeline of inpatient allergy testing during hospitalization as patients may receive multiple doses of second-line antibiotics if testing is delayed. Unfortunately, design of a prospective trial to rule out penicillin allergy may be unethical given that there is sufficient evidence to warrant PST in any patient that would warrant a first-line BLA for infection. The best level of outcome data would have to be obtained from retrospective analysis and longitudinal follow-up of appropriately matched controls.

Longitudinal follow-up would also help quantify (and possibly magnify) the effect of cost. Antibiotic therapy is more costly in patients hospitalized with true penicillin allergy. This is partly due to higher cost per dose of alternative antibiotic and longer duration of treatment.[56] Forrest et al. reported no effect on cost probably owing to the cost benefit solely being calculated at the index admission.[35] However, estimates of annual cost savings are significant and may be more so considering decreased healthcare utilization related to lower risk of health care-related infection and decreased hospital stay.[22,37]

This systematic review and meta-analysis has a number of limitations that are largely related to the quality of studies identified. The level of evidence for such findings was obtained from one case-control study[39] and observational cohort studies.[11,15,32,35,44-46] Unfortunately, we could not identify higher levels of evidence (such as randomized controlled trials) to validate specific risk reductions associated with inpatient PST. Therefore, most studies were retrospective and observational and used convenience sampling. Furthermore, there may have been selection bias with patients on antihistamines and beta-blockers being excluded from PST. Other bias related to indeterminate PST results may have led to result elimination bias.[57] Publication bias may have also precluded reporting of negative clinical outcomes. The results of the pooled-cohort analysis are further limited by the relatively small sample size and convenience sampling in most studies. Further, a number of studies identified were conference abstracts; thus, a rigorous review of the study methodology could not be undertaken. The study findings should be taken in light of the aforementioned limitations. Despite the aforementioned flaws associated with observational studies, the proportion of negative PST was congruent to outpatient data further justifying its use as a safe and reproducible test during hospitalization. However, the review shows that multiple inpatient studies have shown any inpatient intervention addressing a penicillin allergy to be a key component of antibiotic stewardship. This suggests that penicillin skin testing is a valid and reliable tool to evaluate a penicillin allergy when contemplating use of beta-lactam antibiotics. It thus further reaffirms that penicillin skin testing should be strongly recommended as an initial evaluation of penicillin allergy. This recommendation should be reflected in clinical guidelines addressing drug allergy and antibiotic stewardship.[1,20,21,53]

It would be difficult to justify and possibly unethical to conduct a randomized trial where controls are not tested for penicillin allergy given the amount of evidence justifying its use.[5] Given that a timely PST is not readily available in some hospitals,[24] effective inpatient interventions should strive to incorporate guideline-driven protocols that stratify patients most likely to benefit from PST.[10,25] However, it is known that the history of penicillin allergy poorly correlates with patients having a positive PST.[19] Thus, allergists should strive to implement the infrastructure and empower other healthcare providers to safely rule out allergy with PST[36] and follow through by updating the patient's allergy on their medical record.

## 5 | CONCLUSION

Observation data over the past 20 years have shown inpatient penicillin allergy testing to be a validated, safe, and effective method at ruling out penicillin allergy. The rate of negative tests is comparable to outpatient and perioperative data. A history of penicillin allergy should be seriously questioned and proper evaluation should be undertaken in hospitalized patients who require penicillin-based antibiotics.[5] Further longitudinal studies should address the effect of inpatient penicillin allergy testing in relation to clinical outcomes including length of hospital stay, hospital-acquired infection, readmission rate, and cost related to subsequent healthcare utilization. Healthcare providers should be empowered to safely perform PST. This would help risk-stratify patients likely to benefit from early testing and target risk reduction in adverse outcomes related to a documented penicillin allergy.

## CONFLICT OF INTEREST

The authors declare no conflict of interest.

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## SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

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